photons. Although the visible light photons had become weaker and weaker, they never stopped being photons.

What's next on the spectrum? Today, the universe has expanded by a factor of 1,000 from the time photons were set free, and so the cosmic background has, in turn, cooled by a factor of 1,000. All the visible light photons from that epoch have become 1/1,000th as energetic. They're now microwaves, which is where we derive the modern moniker "cosmic microwave background," or CMB for short. Keep this up and fifty billion years from now astrophysicists will be writing about the cosmic radiowave background.

When something glows from being heated, it emits light in all parts of the spectrum, but will always peak somewhere. For household lamps that still use glowing metal filaments, the bulbs all peak in the infrared, which is the single greatest contributor to their inefficiency as a source of visible light. Our senses detect infrared only in the form of warmth on our skin. The LED revolution in advanced lighting technology creates pure visible light without wasting wattage on invisible parts of the spectrum. That's how you can get crazy-sounding sentences like: "7 Watts LED replaces 60 Watts Incandescent" on the packaging.

Being the remnant of something that was once brilliantly aglow, the CMB has the profile we expect of a radiant but cooling object: it peaks in one part of the spectrum but radiates in other parts of the spectrum as well. In this case, besides peaking in microwaves, the CMB also gives off some radio waves and a vanishingly small number of photons of higher energy.

In the mid-twentieth century, the subfield of cosmology—not to be confused with cosmetology—didn't have much data. And where data are sparse, competing ideas abound that are clever and wishful. The existence of the CMB was predicted by the Russian-born American physicist George Gamow and colleagues during the 1940s. The foundation of these ideas came from the 1927 work of the Belgian physicist and priest Georges Lemaître, who is generally recognized as the "father" of big bang cosmology. But it was American physicists Ralph Alpher and Robert Herman who, in 1948, first estimated what the temperature of the cosmic background ought to be. They based their calculations on three pillars: 1) Einstein's 1916 general theory of relativity; 2) Edwin Hubble's 1929 discovery that the universe is expanding; and 3) atomic physics developed in laboratories before and during the Manhattan Project that built the atomic bombs of World War II.